

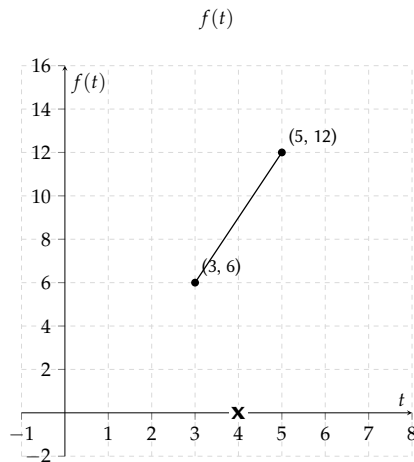
CPE 381 CLASSWORK 04: DISCRETE AND CONTINUOUS REPRESENTATIONS

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 02, 2026 (In-Class)

1. Consider a signal $f(t)$ where $f(3) = 6$, and $f(5) = 12$. What will be the value of signal at $t = 4$, i.e. $f(4)$ if you happen to employ linear interpolation. **(10 Points)**

**Answer**

Starting with two-point line equation form:

$$\frac{f(t) - f(t_1)}{t - t_1} = \frac{f(t_2) - f(t_1)}{t_2 - t_1}$$

$$\frac{f(t) - 6}{t - 3} = \frac{12 - 6}{5 - 3} = \frac{6}{2} = 3$$

$$\Rightarrow \frac{f(t) - 6}{t - 3} = 3$$

So $t = 4$,

$$\frac{f(t) - 6}{4 - 3} = 3$$

$$\frac{f(t) - 6}{1} = 3$$

$$f(t) = 6 + 3 = 9$$

2. What are some non-ideal scenarios you may encounter while dealing with real-world signals? **(5)**

Points)

Answer

Some non-ideal scenario that you may encounter while dealing with real-world signals is

- (a) non-uniform sampling time
- (b) two signals might have different sampling frequency
- (c) signal might be noisy
- (d) signals might miss some values due to packet drop

3. Consider a MATLAB code snippet for the time vector

```
t = 0:0.03:10;
```

- (a) What is the sampling time period T_s ? **(2.5 Points)**
- (b) What is the final time ? **(2.5 Points)**

Answer

$T_s = 0.03$
Final time is 10 unit.